

Supply Chain Analytics — Executive Summary

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Overview

This Power BI model gives leadership unified visibility into supplier performance, delivery bottlenecks, inventory levels, and order fulfillment efficiency — connecting four operational areas that were previously siloed across separate spreadsheets and systems.

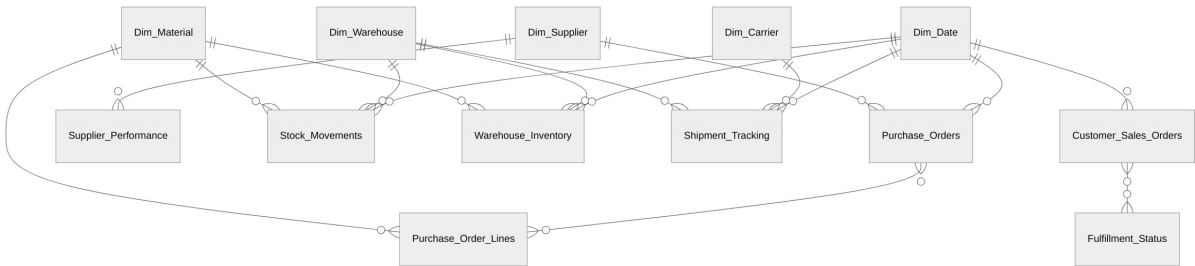
Scope: 12 source datasets, ~180,000 transactional records spanning 2022-2026, built as a governed, secured, analysis-ready semantic model.

Business Problem & Key KPIs

Business question	Answered by
Are suppliers delivering on time and at quality?	On-Time Delivery Rate, Supplier Quality Score, Average Lead Time
Where are the delivery bottlenecks?	Delayed Shipment Rate — the headline metric for this problem
What’s our current inventory position?	Total Quantity On Hand, Stock Movement volume
How efficiently are we fulfilling customer orders?	Order Fulfillment Rate (with MTD/QTD/YTD/YoY tracking)
Is procurement spend trending up or down?	Total Purchase Order Value (full trend/growth tracking)

Model Architecture

Built as a **star schema** — the industry-standard design for fast, reliable BI reporting — with 5 dimension tables (Supplier, Warehouse, Material, Carrier, Date) and 8 fact tables covering purchasing, sales, fulfillment, shipping, and inventory.



Model Relationship Diagram

Every fact table filters through a single shared **Date dimension** (2022-2026), enabling consistent time-based analysis — month-over-month, quarter-over-quarter, and year-over-year — across every business area from one calendar.

Headline Metrics

30 curated measures were built in total, organized into 5 business-area folders plus a focused Time Intelligence layer. The metrics leadership will use most:

Measure	What it tells the business
Total Purchase Order Value (+ YoY/MoM trend)	Total procurement spend and whether it's rising or falling
Delayed Shipment Rate	Share of shipments running behind — the core bottleneck signal
Order Fulfillment Rate (+ YoY/MoM trend)	The single most important measure of operational execution
Avg On-Time Delivery Rate / Quality Score	Supplier reliability at a glance
Total Quantity On Hand	Current inventory position across all warehouses
Cancelled Order Rate	Early warning signal for planning/forecasting problems

(Full catalog of all 30 measures with exact DAX formulas is maintained in the companion Technical Reference document for analysts building or auditing reports.)

Drill-down navigation is built in via four hierarchies: Year → Quarter → Month → Day (calendar), Country → Supplier, Location → Warehouse, and Category → Material — so any report can zoom from a company-wide total down to a single day, supplier, warehouse, or material without extra setup.

Security & Governance

Three role-based access levels control who sees what:

Role	Access
Supply Chain Manager	Full visibility across the entire model
Supplier View	Restricted to a single supplier's own orders and performance data
Warehouse Manager	Restricted to a single warehouse's own inventory, movements, and shipments

Before rolling out Supplier View or Warehouse Manager to real users, each supplier/warehouse record needs to be mapped to an actual login email — currently a placeholder pattern that needs that mapping data to function correctly in production.

Data Quality Approach

Source data contained inconsistent formatting, placeholder values, and conflicting duplicate records. Rather than silently deleting anything ambiguous, every fact table carries a transparent quality flag: clean records are used in all reporting, records with a genuinely missing ID were removed, and records with conflicting data were **preserved and flagged** for manual review — visible in a dedicated Data Quality section of the measures — rather than discarded.

Full data dictionary, all 30 measure formulas, and complete relationship documentation available in the companion Technical Reference document.